

INDUSTRIAL WATER & WASTEWATER ENGINEERING SERIES

DEMINERALIZATION (DM) PLANT

Detailed Engineering & Operations Manual — Expanded Edition

DETAILED ENGINEERING & OPERATIONS MANUAL — EXPANDED EDITION

Water Chemistry Deep-Dives · Process Flow Diagrams · Design, O&M and Project Reference

Industrial Water & Wastewater Engineering Series · Reference Document for Design, O&M and Project Teams

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1. System Profiling & Overview

A Demineralization (DM) Plant removes virtually all dissolved ionic impurities (cations and anions) from water using ion-exchange resins, producing high-purity water essential for boiler feed, power plant steam cycles, pharmaceutical processes, and electronics/semiconductor manufacturing where dissolved solids would cause scaling, corrosion, or product contamination.

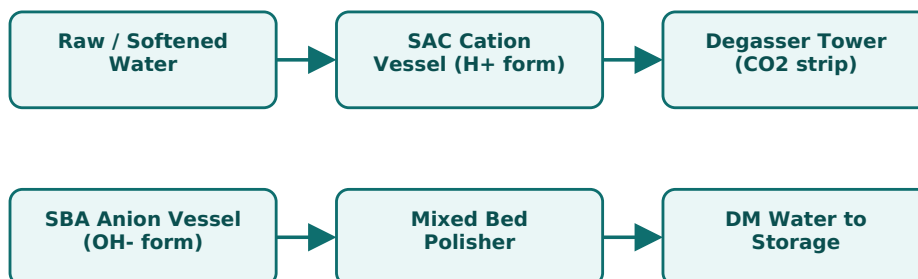


Fig. 1 — Two-Bed + Degasser + Mixed-Bed DM Train (typical high-purity configuration)

1.1 Applications

- High-pressure boiler feed water (power plants, process steam boilers).
- Pharmaceutical purified water pretreatment (upstream of WFI/USP systems).
- Electronics/semiconductor rinse water.
- Laboratory and analytical-grade water production.
- Battery manufacturing and other high-purity process water applications.

1.2 DM Plant Configurations

Configuration	Description	Typical Output Quality
Two-Bed DM (SAC + SBA)	Strong Acid Cation exchanger followed by Strong Base Anion exchanger	Conductivity 10-20 $\mu\text{S}/\text{cm}$
Two-Bed + Mixed Bed	Two-bed DM followed by mixed bed polisher	Conductivity <0.2-1 $\mu\text{S}/\text{cm}$
Degasser-based DM	SAC → Degasser (CO ₂ removal) → SBA → Mixed Bed	Reduces anion resin load, lower regenerant consumption
RO + DM (Hybrid)	RO as primary desalination, DM/Mixed bed as polisher	Lowest regenerant chemical consumption, preferred modern design

2. Feed Water Characterization & Design Basis

Design of resin bed volumes and regeneration frequency is based on feed water total dissolved solids, hardness, alkalinity, silica content, and required daily output volume.

Parameter	Typical Feed Range	Design Relevance
Total Dissolved Solids (TDS)	200-1500 mg/L	Determines total ionic exchange load per cycle
Total Hardness	50-500 mg/L as CaCO ₃	Pre-softening recommended above ~200 mg/L to protect SAC resin
Silica	5-40 mg/L	Critical for high-pressure boiler feed; needs strong base anion resin, degasser ineffective for silica
Free CO ₂ /Alkalinity	10-100 mg/L	Degasser reduces anion resin loading significantly
Turbidity	<5 NTU (post pretreatment)	Must be <1-2 NTU to prevent resin fouling
Total Cations (design basis)	2-15 meq/L	Sizes SAC resin volume and regeneration cycle length
Total Anions incl. Silica (design basis)	2-15 meq/L	Sizes SBA resin volume; silica adds directly to anion exchange load

3. Process Design & Resin Chemistry

- **Strong Acid Cation (SAC) Resin:** Exchanges Ca^{2+} , Mg^{2+} , Na^{+} for H^{+} ions, producing an acidic effluent; regenerated with HCl or H_2SO_4 (typically 4-6% concentration).
- **Strong Base Anion (SBA) Resin:** Exchanges Cl^{-} , SO_4^{2-} , HCO_3^{-} , SiO_2 for OH^{-} ions; regenerated with NaOH (typically 4% concentration), producing demineralized water with H^{+} and OH^{-} combining to form pure H_2O .
- **Mixed Bed Polisher:** Intimate mixture of cation and anion resin in a single vessel gives near-complete ionic removal (conductivity $<0.2 \mu\text{S/cm}$), used as final polishing step.
- **Design flow rate:** Service flow rate typically 16-40 BV/hr (bed volumes per hour) depending on resin type; backwash flow rate 6-8 m/hr to achieve 50-75% bed expansion.
- **Resin Capacity:** SAC resin operating capacity $\sim 1.0\text{-}1.4 \text{ eq/L}$; SBA resin $\sim 0.7\text{-}1.0 \text{ eq/L}$, reduced by 20-30% in practice vs. theoretical to allow for leakage/regeneration efficiency.

4. Pretreatment Systems

Stage	Purpose	Typical Equipment
Multimedia/Pressure Sand Filter	Remove turbidity/suspended solids $<10\text{-}15 \text{ NTU}$ to $<2 \text{ NTU}$	Pressure sand filter with graded media
Activated Carbon Filter	Remove residual chlorine (protects resin from oxidative damage) and organics	Granular activated carbon filter
Softener (if hardness high)	Protect SAC resin from premature exhaustion/fouling by hardness precipitation	Ion-exchange softener (see Softener manual)
Micron Cartridge Filter	Final polishing before resin vessels/RO	5-10 micron cartridge filter housing
Antiscalant Dosing (if RO precedes DM)	Prevent scaling on RO membranes	Dosing pump with antiscalant

5. Regeneration Chemistry & Operation

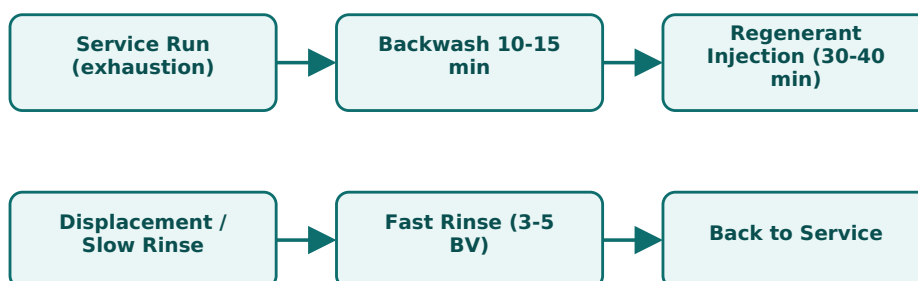


Fig. 2 — DM Regeneration Cycle Sequence

- **Backwash:** Upward flow to declassify and remove trapped solids/broken resin fines, 10-15 min.
- **Regenerant Injection:** Dilute acid (SAC) or caustic (SBA) injected at controlled concentration and slow flow rate (dilute contact $\sim 30\text{-}40 \text{ min}$) to displace exhausted ions from resin.
- **Displacement/Slow Rinse:** Flushes remaining regenerant at injection flow rate.
- **Fast Rinse:** High flow rinse to bring effluent quality to service specification before putting the vessel back into service; rinse water consumption typically 3-5 BV.
- **Regeneration Sequence Automation:** Modern DM plants use PLC-based automatic valve sequencing with conductivity-based end-point detection for consistent regeneration quality.

6. Water & Regeneration Chemistry Deep-Dive

This section expands the underlying ion-exchange chemistry so operators and engineers can diagnose performance from first principles rather than symptom-matching alone.

6.1 SAC Exchange & Regeneration Reactions

Exchange (service): $2R-H + Ca^{2+} \rightarrow R_2-Ca + 2H^+$

$2R-H + Mg^{2+} \rightarrow R_2-Mg + 2H^+$

$R-H + Na^+ \rightarrow R-Na + H^+$

Regeneration (acid): $R_2-Ca + 2HCl \rightarrow 2R-H + CaCl_2$

$R_2-Mg + 2HCl \rightarrow 2R-H + MgCl_2$

Each equivalent of hardness/sodium exchanged releases one equivalent of H^+ into the effluent, which is why SAC effluent pH typically falls to 2.0-3.5 depending on feed TDS. Free mineral acidity (FMA) of the SAC effluent is a direct process-control check on cation exchange completeness.

6.2 SBA Exchange & Regeneration Reactions

Exchange (service): $R-OH + Cl^- \rightarrow R-Cl + OH^-$

$R-OH + HCO_3^- \rightarrow R-HCO_3 + OH^-$

$2R-OH + SO_4^{2-} \rightarrow R_2-SO_4 + 2OH^-$

$R-OH + H_4SiO_4 \rightarrow R-SiO_4H_3 + OH^-$

Regeneration (caustic): $R-Cl + NaOH \rightarrow R-OH + NaCl$

H^+ from the SAC stage and OH^- from the SBA stage recombine ($H^+ + OH^- \rightarrow H_2O$), which is the basis of demineralization. Silica is the weakest-held anion on a standard SBA resin and the first species to leak under exhaustion or cold-regeneration conditions — this is why silica leakage is the earliest and most sensitive indicator of regeneration quality on high-pressure boiler feed duty.

6.3 Regenerant Dose & Capacity Calculation (worked example)

Resin volume required (m³) = Daily ionic load (eq/day) ÷ Operating capacity (eq/m³)

Example — SAC sizing:

Feed hardness + Na = 8 meq/L (0.008 eq/L); Flow = 50 m³/hr; Operating capacity = 1.1 eq/L (1100 eq/m³)

Ionic load = 0.008 eq/L × 50,000 L/hr = 0.4 eq/hr

Run length target = 16 hr → Load per cycle = 6.4 eq

Resin volume = 6.4 eq ÷ 1100 eq/m³ = 0.0058 m³ (scale up per vessel/train count and safety margin)

Regenerant dose is normally expressed as % of theoretical (110-140% of the eq/L capacity figure) to compensate for regeneration inefficiency; over-dosing beyond ~150% gives diminishing capacity return while increasing chemical cost and effluent neutralization load.

6.4 Silica & Leakage Chemistry

Silica exists in water as monosilicic acid (H_4SiO_4), a very weak acid (pK_a ~9.8) that is poorly ionized at neutral pH and therefore only weakly held by SBA resin. Warm caustic regeneration (35-40°C) improves silica desorption kinetics and is standard practice on silica-selective SBA resins for high-pressure boiler duty where outlet silica must be held below 0.02-0.05 mg/L.

6.5 Effluent Neutralization

Spent SAC (acidic, pH 2-3) and SBA (caustic, pH 11-13) regenerant streams are typically combined in a neutralization pit; the acid-base reaction ($H^+ + OH^- \rightarrow H_2O$) is usually sufficient for near-neutral discharge pH, but the combined stream carries high TDS (chloride/sulfate salts) and must still meet local effluent discharge norms for TDS and pH before release or must be routed to the plant ETP.

7. Common Problems & Engineering Solutions

Problem	Root Cause	Engineering Solution
Low treated water quality (high conductivity)	Resin exhaustion, channeling, incomplete regeneration, resin fouling	Verify regeneration dose/contact time, inspect for channeling, resin cleaning/replacement
High silica leakage	Insufficient SBA resin capacity, low regeneration temperature, resin fouling	Use warm caustic regeneration (35-40°C) for silica-selective resin, verify SBA capacity
Shortened run length between regenerations	Higher than design feed TDS, resin fouling by iron/organics	Re-verify feed water quality, consider resin cleaning (iron/organic fouling removal)
Resin fouling by iron	Iron in feed water oxidizing on resin beads	Install iron removal filter upstream, periodic acid/citric cleaning of resin
Excessive regenerant chemical consumption	Over-regeneration practice, resin degradation reducing capacity	Optimize regeneration dose per actual resin capacity test, check resin condition
Resin loss during backwash	Excessive backwash flow rate, broken freeboard/underdrain	Reduce backwash rate, inspect underdrain laterals and freeboard
Pressure drop increase across vessel	Resin fouling, fines accumulation, biological growth	Backwash more frequently, chemical clean resin, consider resin replacement

8. Equipment & Resin Service Life

Equipment/Media	Typical Service Life	Key Life-limiting Factors
SAC Cation Resin	4-6 years (up to 8-10 with good care)	Oxidative attack (chlorine), organic/iron fouling, osmotic shock
SBA Anion Resin	3-5 years	Organic fouling (most sensitive), thermal degradation, oxidative attack
Mixed Bed Resin	3-5 years	Same as above; more sensitive due to fine mixing requirement
FRP/Steel Pressure Vessels	15-20 years	Internal lining degradation, corrosion, rubber lining life (8-10 yrs)
Underdrain/Nozzles	10-15 years	Chemical attack, physical damage during resin change-out
Regeneration Dosing Pumps	5-8 years	Diaphragm wear from acid/caustic exposure
Conductivity/pH Analyzers	3-5 years	Electrode fouling, calibration drift

9. Preventive Service & Maintenance Schedule

Frequency	Activity
Daily	Monitor treated water conductivity/pH; check regenerant tank levels; log service run volume; verify no leaks at valves/pipework
Weekly	Check backwash effectiveness (bed expansion); inspect regenerant dosing accuracy; verify rinse water quality trend
Monthly	Resin bed depth check for loss; test resin sample for capacity (if quality dropping); inspect vessel internals via manway
Quarterly	Iron/organic fouling test on resin; calibrate conductivity/pH analyzers; inspect regenerant dosing pump
Annual	Resin capacity test and replacement planning; internal vessel inspection/lining check; full valve/actuator overhaul; statutory water quality audit

10. Troubleshooting Guide

Symptom	Probable Cause	Corrective Action
Treated conductivity rising steadily	Resin nearing exhaustion	Trigger regeneration; verify against expected run length
Conductivity spike immediately after regeneration	Incomplete rinse, regenerant carryover	Extend rinse cycle, check rinse water conductivity before returning to service
High differential pressure across vessel	Resin fouling/fines, channeling	Backwash, inspect underdrain, consider resin cleaning
Silica breakthrough	Resin exhaustion beyond silica capacity, cold regeneration	Regenerate with warm caustic, verify SBA resin condition
Uneven regenerant distribution	Clogged distributor, poor initial backwash	Inspect and clean regenerant distributor/collector
Frequent premature regeneration	Feed water TDS higher than design, resin capacity loss	Re-test feed water, verify resin capacity, consider resin replacement

11. Project Execution & Engineering Workflow

Phase	Key Deliverables / Activities
Feed Water Analysis & Capacity Study	Complete water analysis, output demand study (m3/day), resin volume and regeneration frequency calculation.
Basic Engineering	Process selection (2-bed/mixed bed/hybrid RO-DM), PFD, equipment sizing, layout concept.
Detailed Engineering	P&ID, vessel GA drawings, regeneration skid design, control philosophy, electrical/instrumentation design.
Procurement	Resin vessels (FRP/rubber-lined MS), resin, dosing systems, PLC/valve automation package.
Civil & Structural	Foundation for vessels, chemical storage bunding (acid/caustic), plant room.
Erection & MEI	Vessel installation, internals fitting, piping, resin loading, control panel wiring.
Pre-commissioning	Hydro test, resin classification backwash, regeneration sequence dry-run.
Commissioning	First regeneration cycles, service run verification against design output quality.
Performance Test	Verify treated water quality (conductivity/silica) at rated flow over sustained run.
Handover & O&M	O&M manual, resin capacity baseline record, chemical safety data sheets, operator training.

12. Standards & Performance Guarantee

- Typical treated water guarantee: conductivity <0.2-1 $\mu\text{S}/\text{cm}$ (with mixed bed), silica <0.02-0.05 mg/L for high-pressure boiler feed applications.
- Regenerant chemical consumption guarantee expressed as kg per m3 of treated water, verified during PGTR.
- Chemical storage and handling must comply with local hazardous chemical storage regulations (acid/caustic bunding, PPE, eyewash stations mandatory near dosing areas).
- Guarantee typically demonstrated over a defined number of complete service-regeneration cycles at rated flow, witnessed by client/consultant with third-party lab verification.

Technical note: figures in this manual are general design guidelines. Final resin volumes, regeneration dosing and operating limits must be established from actual site water analysis and resin manufacturer data.